

Normal Distribution & Z-Scores

MDragon Data Tools - Study Guide

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1. What is a Normal Distribution?

The **normal distribution** (also called Gaussian distribution or bell curve) is a continuous probability distribution that is:

- Symmetric and bell-shaped
- Completely defined by two parameters: mean (μ) and standard deviation (σ)
- The most important distribution in statistics

Real-World Examples

- Heights of adults
- Test scores (IQ, SAT, etc.)
- Measurement errors

- Blood pressure readings
- Manufacturing tolerances

2. Properties and Characteristics

Key Properties

1. **Symmetric** about the mean
2. **Bell-shaped** curve
3. **Mean = Median = Mode** (all at the center)
4. **Total area under curve** = 1 (represents 100% probability)
5. **Continuous** (can take any value)

The Empirical Rule (68-95-99.7 Rule)

For any normal distribution:

- **68%** of data falls within **1 standard deviation** of the mean ($\mu \pm 1\sigma$)
- **95%** of data falls within **2 standard deviations** of the mean ($\mu \pm 2\sigma$)
- **99.7%** of data falls within **3 standard deviations** of the mean ($\mu \pm 3\sigma$)

The diagram illustrates the Empirical Rule for a normal distribution. It shows a horizontal axis with points labeled $\mu - 3\sigma$, $\mu - 2\sigma$, $\mu - 1\sigma$, μ , $\mu + 1\sigma$, $\mu + 2\sigma$, and $\mu + 3\sigma$. Above the axis, three horizontal bars with arrows at their ends represent the percentage of data within specific standard deviation ranges: a short bar between $\mu - 1\sigma$ and $\mu + 1\sigma$ labeled "68%", a medium bar between $\mu - 2\sigma$ and $\mu + 2\sigma$ labeled "95%", and a long bar between $\mu - 3\sigma$ and $\mu + 3\sigma$ labeled "99.7%".

3. The Standard Normal Distribution

The **standard normal distribution** is a special case where:

- Mean (μ) = 0

- Standard deviation (σ) = 1
- Denoted as $Z \sim N(0, 1)$

Why It's Useful

Any normal distribution can be converted to the standard normal distribution using z-scores, allowing us to use standard tables.

4. Z-Scores Explained

What is a Z-Score?

A **z-score** tells you how many standard deviations a value is from the mean.

Z-Score Formula

$$z = (x - \mu) / \sigma$$

where:

x = raw score

μ = population mean

σ = population standard deviation

Interpreting Z-Scores

- **$z = 0$:** Value equals the mean
- **$z = 1$:** Value is 1 standard deviation above the mean
- **$z = -1$:** Value is 1 standard deviation below the mean
- **$z = 2.5$:** Value is 2.5 standard deviations above the mean

Z-Score Sign

- **Positive z :** Value is above the mean

- **Negative z:** Value is below the mean

5. Step-by-Step Examples

Example 1: Calculating a Z-Score

 **Problem:** SAT scores are normally distributed with $\mu = 1000$ and $\sigma = 200$. What is the z-score for a student who scored 1300?

 **Solution:**

1. Identify the values:

- $x = 1300$ (student's score)
- $\mu = 1000$ (mean)
- $\sigma = 200$ (standard deviation)

2. Apply z-score formula:

$$\begin{aligned} z &= (x - \mu) / \sigma \\ &= (1300 - 1000) / 200 \\ &= 300 / 200 \\ &= 1.5 \end{aligned}$$

 **Answer:** $z = 1.5$

Interpretation: The student's score is 1.5 standard deviations above the mean.

Example 2: Finding Probability Using Z-Score

 **Problem:** Using the SAT example above, what percentage of students score below 1300?

 **Solution:**

1. From Example 1, we know $z = 1.5$
2. Look up $z = 1.5$ in standard normal table:

$$P(Z < 1.5) = 0.9332$$

3. Convert to percentage:

$$0.9332 \times 100\% = 93.32\%$$

✓ **Answer:** 93.32% of students score below 1300

Interpretation: A score of 1300 is better than about 93% of test-takers.

Example 3: Finding Probability Between Two Values

 **Problem:** Heights of adult men are normally distributed with $\mu = 70$ inches and $\sigma = 3$ inches. What percentage of men are between 67 and 73 inches tall?

 **Solution:**

1. Calculate z-score for $x = 67$:

$$z_1 = (67 - 70) / 3 = -1$$

2. Calculate z-score for $x = 73$:

$$z_2 = (73 - 70) / 3 = 1$$

3. Find probabilities:

$$P(Z < -1) = 0.1587$$

$$P(Z < 1) = 0.8413$$

4. Calculate difference:

$$\begin{aligned}P(-1 < Z < 1) &= P(Z < 1) - P(Z < -1) \\&= 0.8413 - 0.1587 \\&= 0.6826\end{aligned}$$

✓ **Answer:** 68.26% of men are between 67 and 73 inches tall

Note: This confirms the empirical rule (68% within 1 standard deviation)!

Example 4: Finding a Raw Score from a Percentage

 **Problem:** On a normally distributed test with $\mu = 75$ and $\sigma = 10$, what score represents the 90th percentile?

 **Solution:**

1. Find z-score for 90th percentile:

- Look up 0.90 in z-table (or use calculator)
- $z \approx 1.28$

2. Rearrange z-score formula to solve for x:

$$\begin{aligned}z &= (x - \mu) / \sigma \\x &= \mu + (z \times \sigma) \\x &= 75 + (1.28 \times 10) \\x &= 75 + 12.8 \\x &= 87.8\end{aligned}$$

✓ **Answer:** 87.8 points

Interpretation: To score in the top 10%, you need to score at least 88 points.

Example 5: Using the Empirical Rule

 **Problem:** IQ scores are normally distributed with $\mu = 100$ and $\sigma = 15$. Between what two values do the middle 95% of IQ scores fall?

💡 **Solution:**

1. Apply empirical rule:

- 95% of data falls within $\mu \pm 2\sigma$

2. Calculate boundaries:

$$\text{Lower bound} = \mu - 2\sigma = 100 - 2(15) = 100 - 30 = 70$$

$$\text{Upper bound} = \mu + 2\sigma = 100 + 2(15) = 100 + 30 = 130$$

✅ **Answer:** Between 70 and 130

Interpretation: 95% of people have IQ scores between 70 and 130.

Example 6: Finding Probability in the Upper Tail

📄 **Problem:** Reaction times are normally distributed with $\mu = 200$ ms and $\sigma = 25$ ms. What is the probability of a reaction time greater than 250 ms?

💡 **Solution:**

1. Calculate z-score:

$$z = (250 - 200) / 25 = 2$$

2. Find $P(Z < 2)$:

$$P(Z < 2) = 0.9772$$

3. Use complement rule:

$$\begin{aligned} P(Z > 2) &= 1 - P(Z < 2) \\ &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$

✓ Answer: 0.0228 or 2.28%

Interpretation: About 2.3% of reaction times exceed 250 ms.

Example 7: Unusual Values

 **Problem:** Exam scores have $\mu = 65$ and $\sigma = 12$. A student scores 92. Is this score unusually high?

 **Solution:**

1. Calculate z-score:

$$\begin{aligned} z &= (92 - 65) / 12 \\ &= 27 / 12 \\ &= 2.25 \end{aligned}$$

2. Determine if unusual:

- **Rule of thumb:** $|z| > 2$ is considered unusual
- Since $z = 2.25 > 2$, this is unusual

✓ Answer: Yes, this score is unusually high

Interpretation: The score is more than 2 standard deviations above average, occurring in less than 5% of cases.

6. Using the MDragon Normal Distribution Calculator

Steps:

1. Navigate to **Calculators** → **Normal Distribution**
2. Enter the parameters:
 - **μ (mean):** Average value
 - **σ (standard deviation):** Spread of data

- x : Value of interest

3. The calculator will show:

- Z-score
- $P(X < x)$: Probability below x
- $P(X > x)$: Probability above x
- Visual representation

Tips:

- For "between" problems: Calculate $P(X < \text{upper}) - P(X < \text{lower})$
 - For percentiles: Use inverse calculator feature
 - Verify empirical rule with $\mu \pm 1\sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$
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7. Practice Problems

Problem 1

Weights of apples are normally distributed with $\mu = 150\text{g}$ and $\sigma = 20\text{g}$.

- What is the z-score for an apple weighing 180g?
- What percentage of apples weigh less than 180g?

Problem 2

The average commute time is 35 minutes with $\sigma = 8$ minutes. What percentage of commutes are between 27 and 43 minutes?

Problem 3

Test scores have $\mu = 80$ and $\sigma = 5$. What score represents the 75th percentile?

Problem 4

Battery life is normally distributed with $\mu = 500$ hours and $\sigma = 40$ hours. What is the probability a battery lasts more than 600 hours?

Problem 5

Using the empirical rule, if $\mu = 50$ and $\sigma = 4$, between what two values do:

- a) The middle 68% of values fall?
 - b) The middle 99.7% of values fall?
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Answers to Practice Problems

Problem 1

a) Z-score:

$$z = (180 - 150) / 20 = 1.5$$

b) Percentage below 180g:

$$P(Z < 1.5) = 0.9332 \text{ or } 93.32\%$$

Problem 2

$$z_1 = (27 - 35) / 8 = -1$$

$$z_2 = (43 - 35) / 8 = 1$$

$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826 \text{ or } 68.26\%$$

Problem 3

$$z \text{ for 75th percentile} \approx 0.67$$

$$x = 80 + (0.67 \times 5) = 80 + 3.35 = 83.35$$

Score needed: 83.35 or about 83-84 points

Problem 4

$$z = (600 - 500) / 40 = 2.5$$

$$P(Z > 2.5) = 1 - 0.9938 = 0.0062 \text{ or } 0.62\%$$

Problem 5

a) Middle 68% ($\mu \pm 1\sigma$):

$$\text{Lower: } 50 - 4 = 46$$

$$\text{Upper: } 50 + 4 = 54$$

Between 46 and 54

b) Middle 99.7% ($\mu \pm 3\sigma$):

$$\text{Lower: } 50 - 12 = 38$$

$$\text{Upper: } 50 + 12 = 62$$

Between 38 and 62

Quick Reference

Z-Score Formula

$$z = (x - \mu) / \sigma$$

Finding Raw Score from Z

$$x = \mu + (z \times \sigma)$$

Empirical Rule

- 68% within $\mu \pm 1\sigma$
- 95% within $\mu \pm 2\sigma$
- 99.7% within $\mu \pm 3\sigma$

Common Z-Values

Percentile	Z-Score
90th	1.28
95th	1.645
97.5th	1.96
99th	2.33

Use our Normal Distribution Calculator at MDragon Data Tools for instant calculations!

MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools